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DAY 6

GENERATIVE AI

How AI Creates Text, Images, Audio & Video from a Prompt

A Course Guide by AYE Tech Hub

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1. Introduction

Until 2022, almost every AI system in production was **discriminative** — it could *recognise* things (spam, faces, fraud), *classify* them, and *predict* outcomes. Then, in November 2022, ChatGPT launched and the world changed. Suddenly, AI could **create** — write essays, draw pictures, compose music, generate code, design logos, and even produce video clips from a single sentence. This is **Generative AI**, or **GenAI**, and it has triggered the biggest technology shift since the smartphone.

Generative AI does not just answer questions about existing data — it produces brand-new data that has never existed before. The painting it draws, the article it writes, the voice clip it generates — none of them existed five seconds ago. They are created on demand from a few words of **prompt**. Understanding how this works, what it can and cannot do, and how to use it responsibly is the most valuable skill any engineer can learn in 2026.

Learning Objectives

By the end of this lesson you will be able to:

- Explain the difference between discriminative and generative AI with examples.
- Name the six modalities of generative AI and at least two tools for each.
- Describe how an LLM generates text one token at a time, and how diffusion models create images from noise.
- Write effective prompts using the 5-ingredient formula (role, task, context, format, examples).
- Identify the major risks of GenAI: hallucination, bias, deepfakes, privacy, copyright.
- Choose the right GenAI tool for a given task — writing, coding, design, video, or research.

2. Discriminative vs Generative AI

The whole AI field can be split into two big families. **Discriminative** models (everything you saw in Days 3–5) make decisions *about* existing data — 'is this email spam?', 'whose face is this?', 'will this customer churn?'. **Generative** models do the opposite: they create new data that resembles what they were trained on. Give a generative model a few words and it can produce an image, a song, a story, or a piece of code that did not exist a moment before.

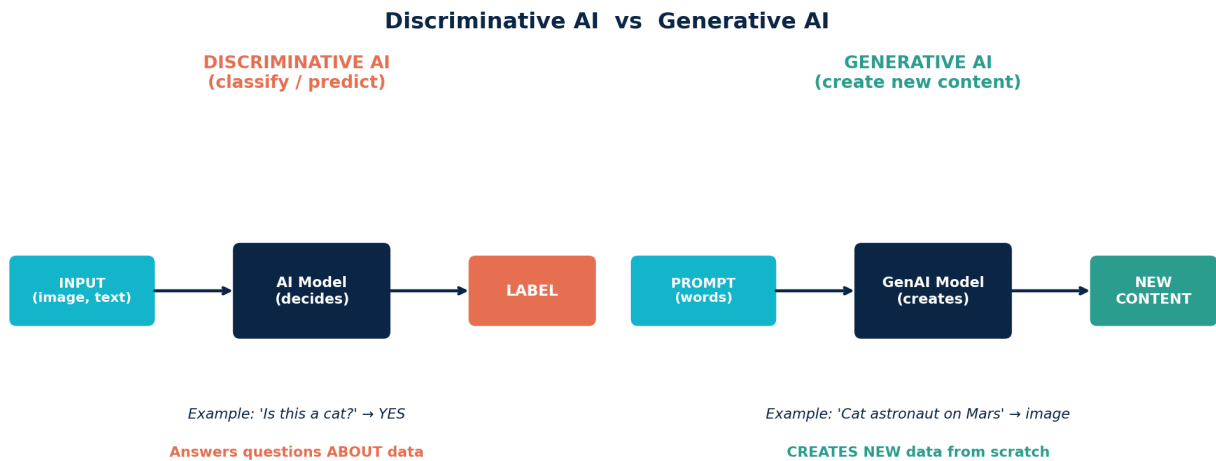


Figure 2.1 — Two AI families. Discriminative AI takes data and outputs a label. Generative AI takes a short prompt and outputs new, original content.

Aspect	Discriminative AI	Generative AI
Question it answers	'What is this?'	'Make me something like this'
Input	Data (image, text, numbers)	Prompt (a short instruction)
Output	A label or number	New content (text, image, audio, video)
Examples	Spam filter, face ID, fraud detection	ChatGPT, DALL-E, Sora, Midjourney
Training	Labelled examples	Massive unlabelled data + reinforcement
Use	Decide / predict	Create / draft / explore

3. Six Modalities of Generative AI

Generative AI is no longer just about text. As of 2026, AI can create across six major **modalities** — text, images, audio, video, code, and 3D objects. Each modality has its leading tools, and the best engineers learn to combine modalities to solve complex real-world problems (for example: generate marketing copy with text AI, illustrate it with image AI, voice it with audio AI, animate it with video AI).

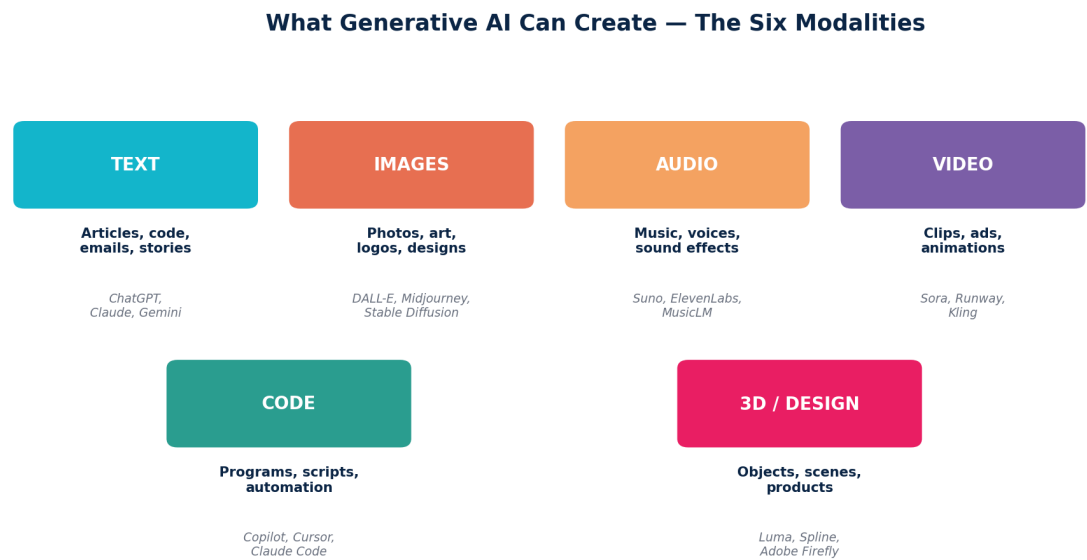


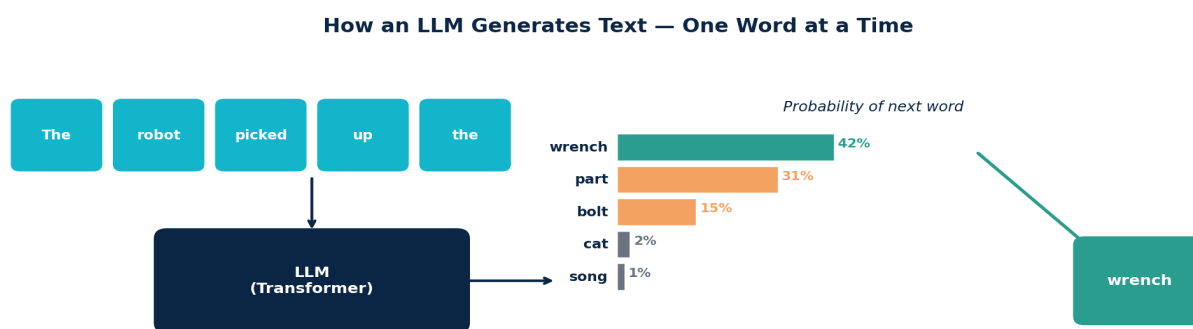
Figure 3.1 — The six modalities of generative AI and the leading tools for each. New tools launch almost every week — the names change, but the categories are stable.

Engineer's Tip

Start with **one modality** and one tool. Most people who try to learn five tools at once give up within a week. Pick text (ChatGPT or Claude) first — it is the most useful across all engineering work. Once you are fluent, add image (Midjourney or DALL-E), then code (Cursor or Claude Code). By the time you reach 30 days you will have mastered the core stack used by 90% of engineers today.

4. How Large Language Models Generate Text

ChatGPT, Claude, and Gemini are all **Large Language Models (LLMs)** — and despite appearances, they do something deceptively simple: they predict **the next word**. Given some input text, the model calculates the probability of every word in its vocabulary being the next word — then picks one (usually the most likely, with some controlled randomness). Add that word to the input. Predict the next word again. Repeat until the answer is complete.



Then add 'wrench' to the input and repeat — this is how the whole sentence is generated.

Figure 4.1 — How an LLM generates 'The robot picked up the wrench'. At each step it ranks every possible next word by probability and selects one.

Tokens, Context Windows & Temperature

Term	Meaning	Why It Matters
Token	A word or piece of a word the model uses	Models charge per token; affects cost and speed
Context window	How many tokens the model can 'see' at once	Larger = can read more documents (now up to 1M+ tokens)
Temperature	Randomness setting (0 = strict, 1 = creative)	Low for factual tasks, higher for creative writing
System prompt	The hidden instruction setting the AI's role	Tells the model HOW to behave for your task
Hallucination	When the model invents facts confidently	Always verify important information

5. How Image Generation Works — Diffusion Models

Image generators like DALL-E 3, Midjourney, and Stable Diffusion use a different approach called a **diffusion model**. The model is trained to do one task: given a noisy image and a text description, slightly reduce the noise so the image looks a little more like the description. To generate a brand-new image, the model starts with pure random noise and applies this denoising step 20–50 times in a row. Each step the image becomes a little less noisy and a little more like your prompt — until a finished image emerges.

How Diffusion Models Create Images — From Noise to Picture

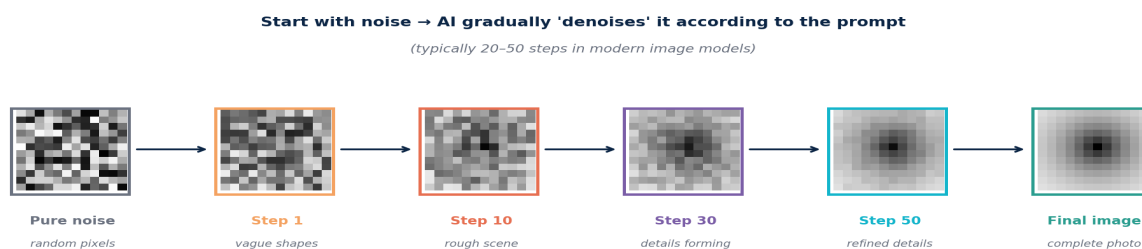


Figure 5.1 — A diffusion model starts with pure noise (left) and gradually denoises it into a finished image (right), guided at every step by the text prompt.

6. Prompt Engineering — The Skill That Multiplies GenAI

The same AI model can produce useless garbage or world-class output, depending entirely on how you ask. **Prompt engineering** is the practical skill of writing instructions that get the best results. It is the difference between an engineer who finds GenAI useful and one who finds it disappointing.

Prompt Engineering — Same AI, Vastly Different Results



The 5 Ingredients of a Great Prompt



Figure 6.1 — Weak prompts give weak results. Strong prompts use five ingredients: Role, Task, Context, Format, and Examples.

7. Practical Use Cases

GenAI is not just a toy — it is now central to how millions of professionals work daily. Engineers use it to accelerate documentation and code; marketers use it to draft campaigns; teachers use it to prepare lessons; designers use it to explore concepts; and students use it to learn. The eight categories below cover roughly 90% of all real-world use today.

Where People Actually Use Generative AI Today

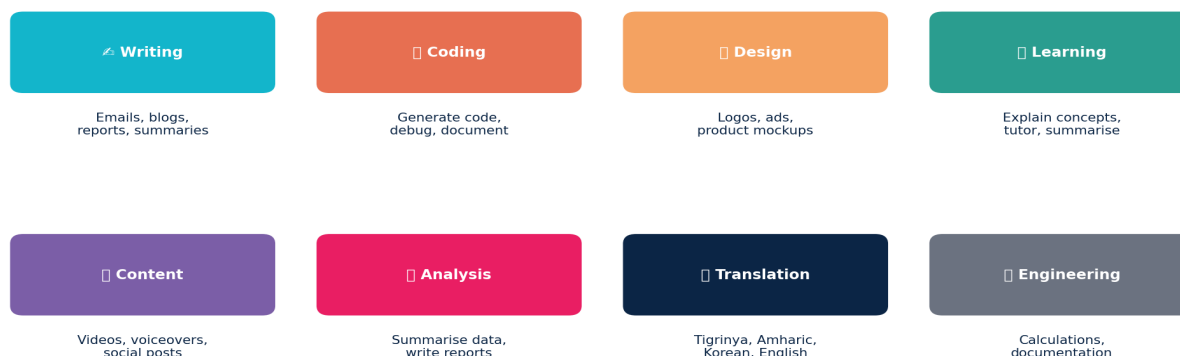


Figure 7.1 — Where people actually use generative AI in 2026. Writing and coding dominate, but the fastest growth is in multimodal (image + video) and language translation.

8. Risks Every User Must Understand

GenAI is powerful — and like any powerful tool it comes with real risks. The five risks below are not theoretical: they have already caused real damage to careers, companies, and reputations. Knowing them is part of being a responsible user.

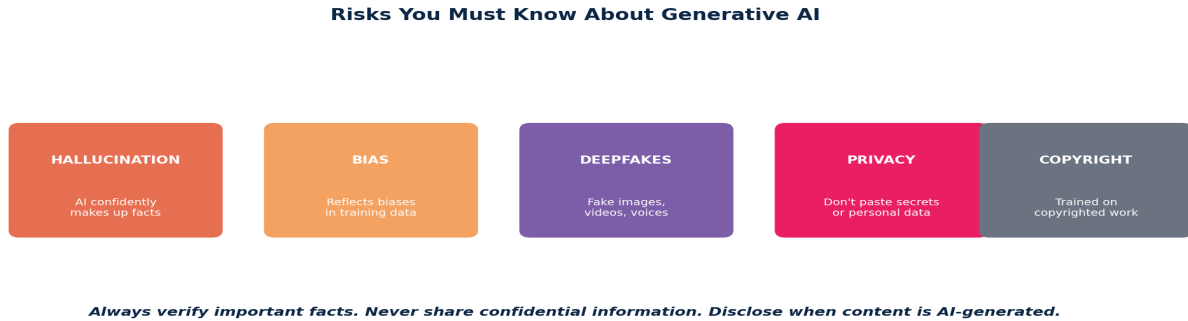


Figure 8.1 — The five most important risks of generative AI. Every user must understand all five before relying on AI output for important decisions.

9. Timeline & Tools You Can Use Today

Generative AI moved from research lab to mainstream tool in just three years. The timeline below shows the major breakthroughs that brought us to where we are today — and where the field is heading next: AI agents that take actions, not just produce content, and reasoning models that can think step-by-step through complex problems.

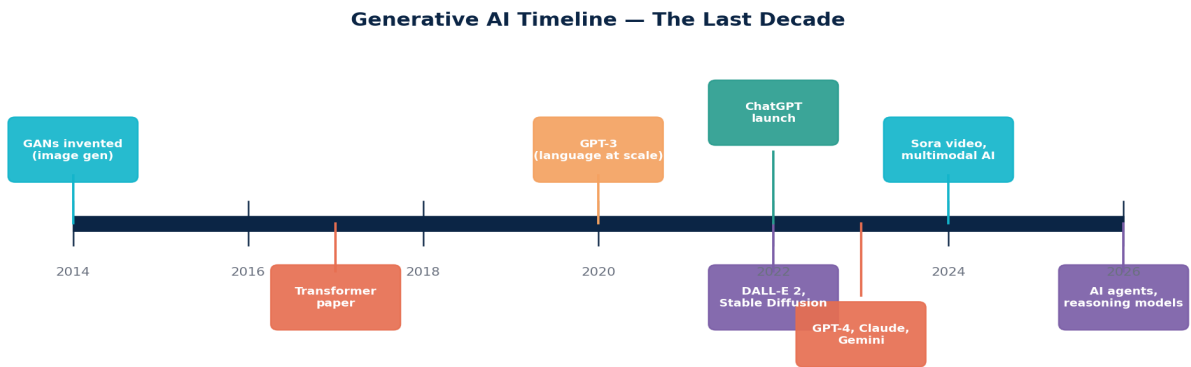


Figure 9.1 — Major generative AI milestones from 2014 to 2026. The pace has accelerated dramatically since ChatGPT launched in November 2022.

Tools to Try Today — Free Tiers Available

Modality	Tool	Best For	Price
Text	Claude (claude.ai)	Long documents, careful reasoning, code	Free tier
Text	ChatGPT (chat.openai.com)	General-purpose, large user base	Free + paid
Text	Google Gemini	Google-product integration, search	Free tier
Image	Microsoft Designer	Logos, social posts, marketing	Free
Image	Stable Diffusion (open source)	Run locally, full control	Free
Code	Claude Code / Cursor	Pair-programming, refactoring, debugging	Free + paid
Audio	ElevenLabs	Voice cloning, narration, dubbing	Free tier
Video	Runway, Kling	Short video clips from text	Free tier



Modality	Tool	Best For	Price
Translate	Naver Papago, Google Translate	Korean, English, multilingual	Free

10. Summary, Self-Check Quiz & Homework

Lesson Recap

- **Generative AI** creates new content — text, images, audio, video, code, 3D — from a short prompt.
- Two AI families: **discriminative** (decide) vs **generative** (create).
- LLMs (ChatGPT, Claude, Gemini) generate text one **token** at a time by predicting the most likely next word.
- Image generators use **diffusion**: start with noise, denoise step-by-step guided by the prompt.
- **Prompt engineering** = the skill of writing instructions that work. The 5 ingredients: role, task, context, format, examples.
- Top use cases: writing, coding, design, learning, content creation, analysis, translation, engineering documentation.
- Risks: **hallucination**, **bias**, **deepfakes**, **privacy leaks**, **copyright**. Always verify facts and never paste secrets.

Self-Check Quiz (answers at the bottom)

1. Name the six modalities of generative AI.
2. How does an LLM produce a full sentence — what is it doing at each step?
3. What technique do most image generators use? Briefly describe how it works.
4. List the five ingredients of a great prompt.
5. What is a hallucination, and how should you handle it?

Answers: 1) Text, images, audio, video, code, 3D/design. 2) Predicts the next word, adds it, repeats until done. 3) Diffusion — start with noise, denoise step-by-step guided by the prompt. 4) Role, task, context, format, examples. 5) AI confidently makes up incorrect facts — always verify important information from independent sources.

Homework Assignment

Pick a real task from your work or studies — drafting an email, summarising a report, writing PLC documentation, planning a class, translating a paragraph — and complete the following:

1. Write a **weak prompt** for it (one short sentence) and note the AI's output.
2. Now write a **strong prompt** using all 5 ingredients (role, task, context, format, examples).
3. Compare both outputs — what changed?
4. Identify **one risk** in the AI output (hallucination? bias? missing detail?).
5. Refine your prompt one more time to remove the risk.

Submit your two prompts + observations to awetgknway@gmail.com with subject *AI Day 6 Homework — Your Name*.

Coming Up Tomorrow — Day 7

Large Language Models (LLMs) in Depth — we go deeper into the tech behind ChatGPT, Claude, and Gemini. You'll learn about transformers, attention, training stages (pretraining, fine-tuning, RLHF), and how to choose the right model for the right job.



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